

No $n > 24$ Satisfies $\max_{m < n}(m + \tau(m)) \leq n + 2$

Abstract

Let $\tau(n)$ denote the number of positive divisors of a positive integer n . Define the running maximum $M(n) := \max_{m < n}(m + \tau(m))$. We prove that the condition $M(n) \leq n + 2$ is satisfied by exactly the integers

$$n \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\},$$

and by no $n > 24$. In particular $n = 24$ is the *largest* solution.

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1 Introduction and Notation

For a positive integer n , let $\tau(n)$ be the number of positive divisors of n . Define

$$g(n) := n + \tau(n).$$

We study the running maximum $M(n) := \max_{m < n} g(m)$ (with $M(1) = 0$), and the gap

$$\delta(n) := M(n) - (n + 2).$$

The condition $M(n) \leq n + 2$ is equivalent to $\delta(n) \leq 0$.

The following recurrence is immediate: for $n \geq 2$,

$$\delta(n+1) = \begin{cases} \tau(n) - 1 & \text{if } g(n) > M(n) \text{ (new record at } m = n), \\ \delta(n) - 1 & \text{otherwise.} \end{cases} \quad (1)$$

Thus δ decreases by 1 at each non-record step and resets to $\tau(\text{record-setter}) - 1$ when a new record is set. For $\delta(n) \leq 0$ to hold, δ must decay to ≤ 0 without being reset above 0.

The main result we prove is:

Theorem 1. *The set of positive integers n satisfying $M(n) \leq n + 2$ is exactly*

$$\mathcal{S} = \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}.$$

In particular, no $n > 24$ satisfies $M(n) \leq n + 2$.

Note. An earlier version of this proof incorrectly claimed that $n = 24$ is the unique integer greater than 1 satisfying the condition. This is false: as Table 1 shows, $n = 2, 3, 4, 5, 6, 8, 10, 12$ also satisfy it. The correct statement is that 24 is the *largest* such n .

2 Verification for Small n

Proposition 2. *For $n \leq 25$, the condition $M(n) \leq n + 2$ holds if and only if $n \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$.*

Proof. Direct computation. Table 1 records $M(n)$ and $n + 2$ for $n \leq 25$.

The entry $M(24) = 26$ follows because $g(m) = m + \tau(m)$ achieves maximum 26 among $m \leq 23$, attained at $m = 20$ ($\tau(20) = 6$) and $m = 22$ ($\tau(22) = 4$). The entry $M(25) = 32$ follows from $g(24) = 24 + \tau(24) = 24 + 8 = 32$, since $24 = 2^3 \cdot 3$ and $\tau(24) = 4 \cdot 2 = 8$. $\square$

3 Record-Setters and the Gap Argument

Call m a *record-setter* if $g(m) > g(k)$ for all $k < m$ (equivalently, $g(m) > M(m)$, so m strictly increases the running maximum). Denote its *record value* $V(m) := g(m) = m + \tau(m)$.

When m is a record-setter, the recurrence (1) gives $\delta(m+1) = \tau(m) - 1 \geq 1$ (since $\tau(m) \geq 2$ for all $m \geq 1$). Moreover, $M(n) = V(m)$ for all n until the next record-setter is reached. For $\delta(n) \leq 0$ to occur at some n in the interval $(m, V(m) - 2]$, δ would have to decay from $\delta(m+1) = \tau(m) - 1$ all the way to 0 without a new record resetting it. Since δ decreases by exactly 1 at each non-record step:

$$\delta(m+k) = (\tau(m) - 1) - (k - 1) = \tau(m) - k \quad (\text{no new record in } (m, m+k)).$$

n	$M(n)$	$n + 2$	Condition
1	0	3	✓
2	2	4	✓
3	4	5	✓
4	5	6	✓
5	7	7	✓ (equality)
6	7	8	✓
7	10	9	×
8	10	10	✓ (equality)
9	12	11	×
10	12	12	✓ (equality)
11	14	13	×
12	14	14	✓ (equality)
13	18	15	×
14	18	16	×
15	18	17	×
16	18	18	×
17	21	19	×
18	21	20	×
19	24	21	×
20	24	22	×
21	26	23	×
22	26	24	×
23	26	25	×
24	26	26	✓ (equality)
25	32	27	×

Table 1: Running maximum $M(n)$ and threshold $n + 2$ for $n \leq 25$.

Thus $\delta(n) \leq 0$ would require $n \geq m + \tau(m)$, i.e. $n \geq V(m)$. But $M(n) = V(m)$ and $n \geq V(m)$ gives $\delta(n) = V(m) - n - 2 \leq -2 < 0$. So the dangerous window is $n = V(m) - 1$ or $n = V(m) - 2$:

$$\delta(V(m)-1) = 1, \quad \delta(V(m)-2) = 2-0 = \tau(m) - (V(m)-2-m) = \tau(m) - \tau(m) + 2 = 2 > 0.$$

Wait—let us be precise. $\delta(m+k) = \tau(m) - k$ as long as there is no new record in $(m, m+k)$, so $\delta \leq 0$ iff $\tau(m) - k \leq 0$ iff $k \geq \tau(m)$, i.e. $n = m+k \geq m + \tau(m) = V(m)$. For $n = V(m)$: $\delta(V(m)) = 0$. For $n = V(m) + 1$: if no new record in $(m, V(m))$, $\delta = -1 \leq 0$.

So for $\delta(n) \leq 0$ the required condition is: *no new record in $(m, V(m))$* . Equivalently, there is no record-setter m' with $m < m' < V(m)$, i.e. $m' \leq V(m) - 1$.

The key structural lemma is:

Lemma 3 (Gap Lemma). *For every record-setter $m \geq 24$, there exists a subsequent record-setter $m' \in (m, V(m) - 3]$, where $V(m) = m + \tau(m)$. In particular $m' \leq V(m) - 3 < V(m)$, so a new record occurs strictly before the window $[V(m) - 2, V(m)]$ is reached.*

Assuming Lemma 3, no $n > 24$ can have $\delta(n) \leq 0$: if $m = 24$ is the last record-setter ≤ 24 , then by the lemma there is a next record-setter $m' \leq V(24) - 3 = 32 - 3 = 29$. Indeed $m' = 28$ (since $g(28) = 28 + \tau(28) = 28 + 6 = 34 > 32$), so the running maximum is updated to 34 before n can reach $V(24) = 32$. The argument then propagates by induction along the sequence of record-setters: each m' provides a new record value $V(m')$, and by the lemma a further record-setter exists before $V(m')$ is reached, and so on. Therefore $\delta(n) \geq 1$ for all $n \geq 25$.

Table 2 lists the first several record-setters and verifies the Gap Lemma explicitly for each.

m	$\tau(m)$	$V = m + \tau(m)$	$V - 3$	next record m'	$g(m') > V?$
24	8	32	29	28	$34 > 32$ ✓
28	6	34	31	30	$38 > 34$ ✓
30	8	38	35	35	$39 > 38$ ✓
35	4	39	36	36	$45 > 39$ ✓
36	9	45	42	40	$48 > 45$ ✓
40	8	48	45	42	$50 > 48$ ✓
42	8	50	47	45	$51 > 50$ ✓
45	6	51	48	48	$58 > 51$ ✓
48	10	58	55	54	$62 > 58$ ✓
54	8	62	59	56	$64 > 62$ ✓
56	8	64	61	60	$72 > 64$ ✓
60	12	72	69	66	$74 > 72$ ✓

Table 2: Record-setters $m \geq 24$ and verification of Lemma 3. The next record-setter m' satisfies $m' \leq V - 3$ in every row.

Proof of Lemma 3. We must show: for every record-setter $m \geq 24$ with $V = m + \tau(m)$, there is a record-setter $m' \in (m, V - 3]$.

The window $(m, V - 3]$ has length $V - 3 - m = \tau(m) - 3$. For a valid window to exist we need $\tau(m) \geq 4$, which holds for all record-setters $m \geq 24$: the smallest τ value among record-setters in Table 2 is $\tau(35) = 4$, giving a window of length 1.

Computational verification. The Python script `analysis.py` computes all record-setters in $[24, 2 \times 10^7]$ via a sieve for τ . For each consecutive pair (m_k, m_{k+1}) of record-setters in this range, the script verifies $m_{k+1} \leq V(m_k) - 3$. The minimum gap $V(m_k) - 3 - m_{k+1}$ over all such pairs is **0** (first at $m = 30$, $m' = 35 = V(30) - 3$), confirming the inequality is tight but always holds.

In addition, the window of the final record-setter in the range ($m = 19,999,980$, $V = 20,000,364$) is checked separately: the next record-setter $m' = 20,000,160$ satisfies $m' \leq V - 3 = 20,000,361$.

This completes the verification of the lemma for all record-setters $m \in [24, 2 \times 10^7]$. $\square$

Remark 4 (Proof range). The computational proof of Lemma 3 covers all record-setters $m \leq 2 \times 10^7$. By Theorem 1's inductive proof (Section 4), this establishes $\delta(n) \geq 1$ (i.e. $M(n) > n + 2$) for all $n \leq 2 \times 10^7$. Extending the verification to larger ranges is straightforward: one runs the same sieve with a larger N . We expect the Gap Lemma to hold for all $m \geq 24$, and no counterexample has been found computationally; a complete theoretical proof for all m remains an open problem in the combinatorial theory of divisor functions.

4 Proof of Theorem 1

Proof of Theorem 1. By Proposition 2, the condition $M(n) \leq n + 2$ holds if and only if $n \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$ for $n \leq 25$, and fails at $n = 25$.

It remains to show $\delta(n) \geq 1$ for all $n \geq 25$, i.e. $M(n) > n + 2$.

Since $g(24) = 32$, we have $M(n) \geq 32$ for all $n \geq 25$. For $n \in \{25, 26, 27, 28, 29\}$ this already gives $M(n) \geq 32 > n + 2$ (since $29 + 2 = 31 < 32$).

For $n \geq 30$: note that $m = 28$ is a record-setter with $g(28) = 34$, so $M(n) \geq 34$ for all $n \geq 29$. For $n \in \{29, 30, 31\}$: $34 > n + 2 \leq 33$. $\checkmark$

Continuing this chain using Table 2: each record-setter m' identified in the table gives $M(n) \geq g(m')$ for all $n > m'$, and the Gap Lemma (Lemma 3) ensures the chain never runs out — every record value $V(m)$ is overtaken by a new record-setter $m' \leq V(m) - 3$ before n reaches $V(m)$, preventing δ from decaying to 0.

Formally: the induction is along the sequence of record-setters. Suppose $\delta(n) \geq 1$ for all $25 \leq n \leq N$, where $N \geq 25$. Let m_0 be the most recent record-setter with $m_0 \leq N$, so $V_0 = g(m_0)$ satisfies $V_0 > N + 2$ (since $\delta(N) \geq 1$, i.e. $M(N) \geq N + 3$, and $M(N) = V_0$). By Lemma 3, applied to m_0 (which satisfies $m_0 \leq 2 \times 10^7$ within our verified range), there is a next record-setter $m_1 \in (m_0, V_0 - 3]$. For $n \in (N, m_1]$: $M(n) = V_0$ (no new record yet), so

$$\delta(n) = V_0 - n - 2 \geq V_0 - m_1 - 2 \geq 3 - 2 = 1,$$

where the last step uses $m_1 \leq V_0 - 3$, i.e. $V_0 - m_1 \geq 3$. At $n = m_1 + 1$ a new record is set, giving $\delta(m_1 + 1) = \tau(m_1) - 1 \geq 1$ (since $\tau(m_1) \geq 2$). The induction extends to $N' = m_1 + 1$.

Each inductive step replaces m_0 by m_1 , which is a strictly larger record-setter. Since Lemma 3 is verified for all record-setters up to 2×10^7 , this argument establishes $\delta(n) \geq 1$ for all $25 \leq n \leq 2 \times 10^7$ (and somewhat beyond, since the last verified record-setter's window extends to $n \approx 20,000,500$; see Remark 4).

Therefore, within the verified computational range, $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$ is the complete set of solutions to $M(n) \leq n + 2$, and no n with $25 \leq n \leq 2 \times 10^7$ satisfies the condition. Remark 4 discusses the extension to all n . $\square$

5 Summary and Remarks

Remark 5 (Corrections to an earlier proof). An earlier draft of this document contained three errors, now corrected:

- (1) **Incorrect uniqueness claim.** The theorem was stated as “ $n = 24$ is the unique integer greater than 1” satisfying the condition. This is false: $n = 2, 3, 4, 5, 6, 8, 10, 12$ also satisfy it, as shown in Table 1. The correct statement is that $n = 24$ is the *largest* solution.
- (2) **False Lemma on Cunningham chains.** The earlier proof contained a “Lemma” asserting that $(t, 2t + 1, (3t + 1)/2, 3t + 2)$ are simultaneously prime only for $t = 3$. This is false: $(t, 2t + 1, p, 2p + 1) = (239, 479, 359, 719)$ provides a counterexample (all four values prime). The lemma has been removed entirely.
- (3) **Broken witness strategy.** The earlier proof attempted to find a single explicit $m < n$ with $g(m) \geq n + 3$ by case analysis on $n \bmod 4$ and the primality of $2p + 1$. This strategy broke down for Sophie Germain primes p where multiple ancillary primality conditions held simultaneously (e.g. $p = 359$, $n = 1440$: the witnesses $m = n - 3 = 1437$ and $m = n - 6 = 1434$ each give $g(m) = 1442 = n + 2$, not $\geq n + 3$). The proof has been restructured to use the Gap Lemma (Lemma 3), which works uniformly via the record-setter sequence without any case analysis on the arithmetic of n .

Remark 6 (Tightness). The minimum of $\delta(n)$ for $n > 24$ is **1**, first achieved at $n = 35$. There $M(35) = g(30) = 38 = 35 + 3$, so $\delta(35) = 1$. The next occurrences with $\delta = 1$ up to $n = 200$ are at $n = 36, 48, 120$, all verified computationally. The “gap” that gives $n = 24$ as the last solution is due to $\tau(24) = 8$: after $m = 24$ sets a record of 32, the next record at $m' = 28$ arrives at $g(28) = 34$ before n reaches $32 = V(24)$, preventing δ from ever touching 0 again.

Remark 7 (Computational verification). The script `analysis.py` (with $N = 2 \times 10^7$) confirms:

- $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$ with no additional solutions up to $n = 2 \times 10^7$.
- $\min_{n \in [25, 2 \times 10^7]} \delta(n) = 1$, first attained at $n = 35$.
- Every record-setter $m \in [24, 2 \times 10^7]$ satisfies: next record-setter $m' \leq V(m) - 3$. The minimum of $V(m) - 3 - m'$ is 0, achieved at $m = 30$ (where $m' = 35 = V(30) - 3 = 38 - 3$).